Liver disease, especially **cirrhosis**, causes oedema through several linked mechanisms. The result is peripheral oedema (legs, sacrum) and **ascites** (fluid in the peritoneal cavity).

1. Hypoalbuminaemia: reduced plasma oncotic pressure

- The liver makes albumin, the main plasma protein that holds fluid in the vessels through oncotic pressure.
- Damaged hepatocytes make less albumin, so plasma oncotic pressure falls.
- By Starling forces, fluid moves out of capillaries into the interstitial space, causing oedema.
- This is a contributing factor, not the main driver of ascites.

2. Portal hypertension: raised hydrostatic pressure

- Fibrosis and regenerative nodules in cirrhosis distort the liver architecture and obstruct portal blood flow, raising pressure in the portal vein and splanchnic capillaries.
- Increased hydrostatic pressure in the splanchnic capillaries pushes fluid into the peritoneal cavity, forming **ascites**.
- Raised hepatic sinusoidal pressure also increases lymph production, which overwhelms the lymphatic drainage capacity, and fluid weeps from the liver surface.
- This is the main driver of ascites.

3. Splanchnic vasodilation and secondary hyperaldosteronism

This explains why the body retains salt and water.

- Portal hypertension increases nitric oxide (and other vasodilators) in the splanchnic circulation, causing marked **arterial vasodilation**.
- Effective arterial blood volume falls, so the kidneys sense hypovolaemia even though total body fluid is high.
- Compensatory systems activate:
    - **Renin-angiotensin-aldosterone system (RAAS):** aldosterone increases sodium and water reabsorption in the distal tubule.
    - **Sympathetic nervous system** activation.
    - **ADH (vasopressin):** water retention, which can cause dilutional hyponatraemia.

- Also, the failing liver cannot break down aldosterone, adding to the effect (a minor contribution).
- The retained sodium and water leak into the peritoneal cavity and interstitium, perpetuating ascites and oedema.

4. Hepatic venous outflow obstruction

In Budd-Chiari syndrome (hepatic vein thrombosis), hepatic sinusoidal pressure rises sharply, producing ascites by the same hydrostatic mechanism.

Summary of the sequence

Cirrhosis → portal hypertension + hypoalbuminaemia → splanchnic vasodilation → fall in effective circulating volume → RAAS, sympathetic and ADH activation → sodium and water retention → ascites and peripheral oedema.

Clinical pointers

- Ascites typically appears first, with leg oedema following as albumin falls and fluid retention worsens.
- Management targets these mechanisms: sodium restriction, **spironolactone** (aldosterone antagonist, often with furosemide), paracentesis with albumin infusion for large ascites, and treating the underlying liver disease.
- Complications include spontaneous bacterial peritonitis and hepatorenal syndrome (severe renal vasoconstriction from the same circulatory disturbance).